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Probabilistic Wind Power Forecast-Error Generation using a Conditional GAN

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Abstract

Reliable operation of modern power grids requires robust methods for capturing uncertainty in wind power predictions. This work introduces a Conditional Generative Adversarial Network (CGAN) implemented from the ground up in PyTorch to synthesise realistic 24-hour Forecast Error (FE) scenarios simultaneously across 36 spatially-correlated wind farms drawn from the NREL WIND Toolkit dataset. Unlike conventional CGAN designs that flatten the conditioning input, this work introduces a **three-tier structured conditioning pathway** that exposes every output cell (hour h , farm f) to (i) a global Forecast Power (FP) summary, (ii) per-hour and per-farm slice embeddings, and (iii) the local FP scalar, thereby eliminating the variance-collapse failure mode that affects flat-conditioning approaches. The model is trained using Binary Cross-Entropy with Spectral Normalisation, Two Time-scale Update Rule (TTUR), Exponential Moving Average (EMA) of generator weights, and one-sided label smoothing. A per-cell decoder shared across all $24 \times 36 = 864$ output cells preserves parameter efficiency while allowing each cell to receive its own structured condition. The trained CGAN is rigorously evaluated using 20 independent ensembles of 1,000 generated scenarios benchmarked against approximately 439 held-out test days, with a comprehensive suite of probabilistic metrics covering distributional fidelity (Wasserstein-1, CRPS, QQ R^2 , CDF), reliability (90% prediction-interval coverage), and temporal structure (inter-hour correlation). The model achieves a best Wasserstein-1 distance of 0.029, a mean CRPS of 0.055, and a CRPS standard deviation of 2.3×10^{-5} across 20 independent ensembles, confirming the absence of mode collapse. The results highlight both the strengths of the structured-conditioning approach and the residual challenges of CGAN-based scenario forecasting under limited training data. Source code is accessible at: Google Drive Folder.

Keywords: Conditional GAN, structured conditioning, wind energy forecasting, probabilistic forecasting, forecast error, scenario generation, spectral normalisation.

1 Introduction

The increasing integration of renewable energy sources, particularly wind power, has intensified the need for accurate and reliable forecasting systems. Because wind conditions are inherently stochastic, predicting wind power generation introduces substantial uncertainty into the operation of modern power grids. Traditional deterministic forecasts return only a single expected trajectory and provide no information about spread, leaving system operators without the probabilistic context required for effective reserve scheduling, unit commitment, or financial hedging.

Classical statistical forecasting techniques such as ARIMA, SARIMA, and quantile regression have been widely employed for renewable-energy time-series prediction, but they often struggle to handle the non-linear and non-stationary behaviour of real-world wind data. Deep learning approaches, in contrast, are capable of capturing complex dependencies directly from historical observations without explicit distributional assumptions, and have therefore become the dominant paradigm in recent renewable-energy forecasting literature.

Generative Adversarial Networks (GANs) provide a particularly attractive framework for sampling from high-dimensional data distributions. The conditional extension of the GAN framework augments the noise input with an auxiliary conditioning signal — in this study, the 24-hour FP profile across 36 farms — so that synthesised samples remain statistically coherent with the observed operating state.

This project develops a CGAN designed to produce 24-hour FE scenarios *simultaneously* for all

36 wind farms in the NREL WIND Toolkit dataset, conditioned on the corresponding 24-hour FP profile across the same farms. A central contribution is the *three-tier structured conditioning pathway*, introduced specifically to address the variance-collapse failure mode observed in earlier flat-conditioning iterations. The model is subsequently evaluated through a structured battery of proper scoring rules and diagnostic visualisations that illuminate both strengths and residual failure modes.

The principal contributions of this study are: (i) a thoroughly documented structured-conditioning CGAN architecture and training protocol adapted to multi-farm wind FE scenario synthesis; (ii) a multi-dimensional evaluation methodology that addresses distributional fidelity, calibration quality, spatial coupling across 36 farms, and temporal structure within each daily profile; and (iii) an objective examination of the persisting shortcomings alongside concrete recommendations for mitigation.

2 Objectives

The project aims to explore how deep generative models can be used to improve wind power forecasting in renewable energy systems. Specifically, the objectives are:

- To build a Conditional Generative Adversarial Network (CGAN) capable of generating realistic and diverse wind power forecast-error scenarios across 36 wind farms.
- To investigate how effectively deep learning models can capture complex and non-linear wind patterns compared with traditional statistical forecasting methods.
- To design and validate a *structured conditioning pathway* that overcomes the variance-collapse failure mode commonly observed in flat-conditioning CGAN architectures.
- To train the model using historical wind data so that it can learn both the temporal variations within a day and the spatial coupling across farms.
- To evaluate the accuracy and reliability of the generated scenarios using a comprehensive suite of probabilistic metrics, comparing them against held-out real-world wind data.

3 Related Work

Probabilistic wind power forecasting has been pursued through diverse approaches, encompassing parametric distributional models, quantile regression, kernel density-based methods, and more recently deep generative architectures. Normalising flows and variational autoencoders provide tractable likelihoods but may encounter difficulties modelling sharply multimodal distributions. GANs circumvent this limitation by framing distribution learning as an adversarial optimisation, which has proven effective for the heavy-tailed and conditionally heteroscedastic FE distributions characteristic of wind power.

Prior work has applied GAN-based methods across energy domains, including deep learning strategies for wind forecasting and Wasserstein-GAN applications to solar irradiance scenario synthesis. Wind-specific CGAN architectures conditioned on numerical weather prediction variables have also appeared in the literature, although most existing studies either aggregate across farms or restrict analysis to a single installation.

From a training-stability perspective, spectral normalisation constrains the discriminator’s Lipschitz constant without incurring the computational overhead associated with gradient-penalty

methods. The TTUR regime exploits deliberate learning-rate asymmetry to promote convergence, and EMA smoothing of generator weights yields more stable generative output than using raw instantaneous parameters.

For probabilistic evaluation, the Continuous Ranked Probability Score (CRPS) constitutes the standard proper scoring rule for scalar predictive distributions, integrating sharpness and calibration into a single diagnostic quantity. Wasserstein-1 distance furnishes a geometrically interpretable measure of discrepancy between empirical distributions even when their supports do not overlap. The Adam optimiser underpins parameter updates throughout training, with modified momentum settings tuned for the non-stationary dynamics of adversarial learning.

4 Problem Formulation

Let $f \in \{0, 1, \dots, 35\}$ denote a wind-farm index and d a calendar day. For day d :

- $\mathbf{P}^{(d)} \in \mathbb{R}^{24 \times 36}$: the 24-hour Forecast Power matrix across all 36 farms (the conditioning variable).
- $\mathbf{E}^{(d)} \in \mathbb{R}^{24 \times 36}$: the 24-hour Forecast Error matrix across all 36 farms (the generation target).

The objective is to train a conditional generator

$$G : \mathbb{R}^{N_z} \times \mathbb{R}^{24 \times 36} \longrightarrow \mathbb{R}^{24 \times 36}$$

such that, for a latent vector $z \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, the distribution of $G(z, \mathbf{P})$ approximates the true conditional distribution $p(\mathbf{E} \mid \mathbf{P})$. At inference time, drawing S independent noise realisations under a fixed FP profile produces an S -member scenario ensemble suitable for downstream stochastic optimisation or risk-assessment tasks.

5 Dataset and Preprocessing

5.1 Data Description

The NREL WIND Toolkit dataset `wind_gan.csv` provides hourly wind power records across 36 spatially-correlated wind farms over a six-year period, organised into 72 columns of hourly measurements. Columns 0–35 hold FE values for the corresponding wind farms, while columns 36–71 contain the associated FP values. Each row records one hour of observation, amounting to approximately 52,560 rows in total.

5.2 Sample Construction

Unlike conventional approaches that treat each (farm, day) pair as an independent sample, this work groups all 36 farms within a single day into one composite sample. Specifically, for day d :

$$\begin{aligned} \mathbf{P}^{(d)} &= \mathbf{P}_{\text{raw}}[24d : 24(d+1), \text{ cols } 36\text{--}71] \in \mathbb{R}^{24 \times 36}, \\ \mathbf{E}^{(d)} &= \mathbf{E}_{\text{raw}}[24d : 24(d+1), \text{ cols } 0\text{--}35] \in \mathbb{R}^{24 \times 36}. \end{aligned}$$

Each sample is therefore of shape $(24, 72)$. For the network, both matrices are flattened in row-major order (index = $h \times 36 + f$) into 864-dimensional vectors:

$$\mathbf{e}_{\text{flat}} \in \mathbb{R}^{864}, \quad \mathbf{p}_{\text{flat}} \in \mathbb{R}^{864}.$$

This day-level grouping preserves cross-farm spatial coupling *within* each scenario — a property that single-farm sample formulations cannot capture.

5.3 Train/Test Split and Feature Scaling

Days are partitioned chronologically: the leading 80% form the training corpus and the trailing 20% constitute a held-out evaluation set. From a six-year window of 2,190 days, this yields approximately 1,752 training days and 438 test days. A global Min-Max scaler fitted exclusively on training data maps all values linearly into $[0, 1]$:

$$x_{\text{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}.$$

Separate scalers are fitted for FE and FP. A single $(x_{\min}, x_{\max})$ pair is applied across all hours, days, and farms within each variable, rather than computing per-hour or per-farm statistics. Per-hour normalisation would force each hour’s maximum to unity, erasing the amplitude variation that the generator must reproduce. Both scalers are saved to disk for reuse during evaluation.

6 Model Architecture

6.1 Structured Conditioning Pathway

A central contribution of this work is the three-tier structured conditioning pathway. In a flat-conditioning baseline, the entire FP day (864 numbers) is provided to the generator as a single vector. To produce one FE value at cell (h, f) , the network must extract the relevant signal from 864 mostly-irrelevant entries, which the network learns to “solve” by collapsing to a near-deterministic mean given the condition — producing narrow, low-variance generated CDFs.

The structured pathway eliminates this incentive by exposing every output cell (h, f) to FP information *organised by relevance*:

- **Tier 1 — Global FP summary.** A 3-layer MLP compresses the 864-dim flattened FP into a 256-dim global representation that captures the overall FP signature of the day.
- **Tier 2 — Per-hour and per-farm slices.** The hour-slice projector maps each 36-dim row $\mathbf{P}[h, :]$ to a 64-dim embedding (“what every farm forecasts at hour h ”). The farm-slice projector maps each 24-dim column $\mathbf{P}[:, f]$ to a 64-dim embedding (“what farm f forecasts across the day”).
- **Tier 3 — Local FP scalar.** The exact scalar $\mathbf{P}[h, f]$ is fed directly into the per-cell decoder for that cell.

For each output cell, the network thus receives a concatenation of: global FP summary (256-d), hour-slice embedding (64-d), farm-slice embedding (64-d), local FP scalar (1-d), per-cell coupled noise (32-d), and a global noise summary (64-d) — yielding a 481-dimensional per-cell condition vector.

6.2 Generator

The generator G accepts a latent vector $z \in \mathbb{R}^{128}$ and the structured FP condition. The Tier-1 global summary, Tier-2 hour and farm projections, and Tier-3 local scalar are computed once per sample. Per-cell noise is generated through a multiplicative coupling of an hour-noise tensor (B, 24, 32) and a farm-noise tensor (B, 36, 32), giving every cell its own noise channel at $\mathcal{O}(N_{\text{hours}} + N_{\text{farms}})$ cost rather than $\mathcal{O}(N_{\text{hours}} \times N_{\text{farms}})$.

A shared per-cell decoder, an MLP with structure $481 \rightarrow 256 \rightarrow 128 \rightarrow 1$ using LayerNorm and LeakyReLU activations, is then applied to all $24 \times 36 = 864$ cells in parallel. A learnable per-farm bias is added before a final sigmoid maps each output to $[0, 1]$, consistent with the Min-Max normalisation. The complete generator architecture is summarised in Table 1.

Table 1: Generator (G) architecture.

Layer	Specification
Input	$z \in \mathbb{R}^{128} + \text{structured FP condition}$
Tier 1 – Global FP MLP	Linear $864 \rightarrow 512 \rightarrow 256 \rightarrow 256$ (LayerNorm + LeakyReLU)
Tier 2 – Hour-slice projector	Linear $36 \rightarrow 128 \rightarrow 64$ (per hour)
Tier 2 – Farm-slice projector	Linear $24 \rightarrow 128 \rightarrow 64$ (per farm)
Hour-noise MLP	Linear $128 \rightarrow 256 \rightarrow (24 \times 32)$
Farm-noise MLP	Linear $128 \rightarrow 256 \rightarrow (36 \times 32)$
Global-noise projection	Linear $128 \rightarrow 64$
Per-cell decoder (shared)	$481 \rightarrow 256 \rightarrow 128 \rightarrow 1$ (LayerNorm + LeakyReLU)
Output	+ per-farm bias, Sigmoid $\rightarrow \text{FE} \in \mathbb{R}^{864}$

6.3 Discriminator

The discriminator D receives the concatenation of either a generated or real FE field with the FP condition, forming a 1728-dimensional input. Four spectral-normalised linear layers progressively reduce dimensionality before producing a raw logit. Spectral normalisation is applied to every linear layer in D for Lipschitz stability without the additional forward passes required by gradient-penalty formulations. Dropout regularisation is inserted between intermediate layers at rates 0.4, 0.4, and 0.3 to prevent memorisation of the modest training set.

Table 2: Discriminator (D) architecture.

Layer	Specification
Input	[FE, FP] concatenated $\rightarrow 1728$
Layer 1	SN(Linear $1728 \rightarrow 512$) + LeakyReLU + Dropout 0.4
Layer 2	SN(Linear $512 \rightarrow 512$) + LeakyReLU + Dropout 0.4
Layer 3	SN(Linear $512 \rightarrow 256$) + LeakyReLU + Dropout 0.3
Layer 4	SN(Linear $256 \rightarrow 128$) + LeakyReLU
Output	SN(Linear $128 \rightarrow 1$) $\rightarrow$ raw logit

7 Training Methodology

7.1 Loss Function and Label Smoothing

Binary Cross-Entropy with logits serves as the adversarial objective:

$$\mathcal{L}_D = \text{BCE}(D(\mathbf{e}_{\text{real}}, \mathbf{p}), 0.9) + \text{BCE}(D(\mathbf{e}_{\text{fake}}, \mathbf{p}), 0), \quad (1)$$

$$\mathcal{L}_G = \text{BCE}(D(\mathbf{e}_{\text{fake}}, \mathbf{p}), 1). \quad (2)$$

Real training samples receive a soft target of 0.9 rather than the hard binary label 1.0. This one-sided label-smoothing strategy reduces overconfidence in the discriminator, which in practice yields smoother gradient feedback to the generator and diminishes the tendency toward mode dropping.

7.2 Two Time-scale Update Rule

The discriminator is assigned a learning rate four times larger than that of the generator: $\text{LR}_D = 2 \times 10^{-4}$, $\text{LR}_G = 5 \times 10^{-5}$. This intentional asymmetry reflects the theoretical observation that a faster-converging discriminator supplies more accurate gradient information to the generator during joint training. Both optimisers use Adam with $\beta_1 = 0.5$, $\beta_2 = 0.999$; the lowered β_1 reduces gradient momentum, a modification beneficial in non-stationary adversarial dynamics.

7.3 Spectral Normalisation

Each linear weight W in D is normalised by its largest singular value:

$$W_{\text{SN}} = \frac{W}{\sigma(W)},$$

where $\sigma(W)$ is the largest singular value of W . This bounds the Lipschitz constant of D ($\|D\|_{\text{Lip}} \leq 1$) and stabilises training.

7.4 Exponential Moving Average of Generator Weights

An EMA shadow copy of the generator is maintained throughout training:

$$\theta_{\text{EMA}} \leftarrow \beta \theta_{\text{EMA}} + (1 - \beta) \theta_G, \quad \beta = 0.999. \quad (3)$$

All evaluations and model checkpoints employ the EMA parameters rather than the raw instantaneous weights. The smoothed model produces markedly more consistent samples, particularly in the later stages of training when the instantaneous weights oscillate around a local Nash equilibrium.

7.5 Wasserstein-1 Distance for Checkpoint Selection

The Wasserstein-1 (earth-mover) distance is used as the convergence signal:

$$W_1(P, Q) = \frac{1}{n} \sum_{i=1}^n |\text{sort}(P)_i - \text{sort}(Q)_i|. \quad (4)$$

Computed every 25 epochs across a randomly sampled subset of (farm, hour) test cells, W_1 governs both checkpoint selection and early stopping.

7.6 Learning-Rate Scheduling and Regularisation

Cosine annealing reduces each learning rate to 5% of its initial value across the maximum training horizon. Gradient norms are clipped to a maximum of 5.0 to prevent parameter explosion. Mixed-precision (AMP) arithmetic is used on GPU hardware. Early stopping triggers after 4 consecutive evaluations without improvement on W_1 , with a minimum-epoch floor of 200.

7.7 Hyperparameters

The complete set of training hyperparameters is summarised in Table 3.

Table 3: Training hyperparameters.

Setting	Value
Sample shape	24×72 (per day)
Batch size	64
Maximum epochs	800 (with early stopping)
Minimum epochs before early stop	200
Early-stop patience	4 evaluations (≈ 100 epochs)
Evaluation frequency	Every 25 epochs
Optimiser	Adam, $\beta = (0.5, 0.999)$
Learning rate G	5×10^{-5}
Learning rate D	2×10^{-4} (TTUR ratio 4:1)
LR schedule	Cosine annealing $\rightarrow$ 5% of base LR
Gradient clipping	5.0 (G and D)
Label smoothing (real)	0.9
EMA decay (G weights)	0.999
D dropouts	0.4/0.4/0.3
Generator regularisation	LayerNorm throughout
AMP (mixed precision)	Enabled on CUDA
Random seed	42

8 Evaluation Metrics

A multi-dimensional assessment framework examines distinct facets of generative quality. Rather than reporting metrics from a single generation pass, evaluation is performed across $N_{\text{sets}} = 20$ independent ensembles of $N_{\text{gen}} = 1,000$ scenarios per test day — yielding approximately $20 \times 1,000 \times 439 \approx 8.78$ million scenario draws in total — so that cross-set stability can itself be quantified.

Wasserstein-1 Distance. W_1 quantifies the earth-mover cost required to transport one empirical distribution onto another at each (farm, hour) cell. Unlike KL divergence, W_1 remains finite and geometrically interpretable even when the two distributions share no overlapping support — a routine situation when wind output approaches zero or rated capacity. It serves as the primary criterion for checkpoint selection and early stopping.

Continuous Ranked Probability Score (CRPS). CRPS is a strictly proper scoring rule rewarding both sharpness and calibration:

$$\text{CRPS}(F, y) = \int_{-\infty}^{\infty} (F(x) - \mathbb{I}[x \geq y])^2 dx. \quad (5)$$

A value of zero corresponds to a perfect probabilistic forecast. CRPS is averaged across all farms and all 24 hours to produce a global scalar summary, and its standard deviation across the 20 evaluation sets is reported to confirm cross-run stability.

Quantile–Quantile Plot and R^2 . Empirical quantiles of generated samples are matched against real FE quantiles from FP-similar test days. An R^2 value near 1.0 signals accurate reproduction of the conditional marginal distribution. The metric is computed independently for each of the 24 hours, and the mean is reported.

CDF Comparison. A per-hour, per-farm visual diagnostic. A well-calibrated generator should produce a CDF curve that follows the conditional real CDF (constructed from FP-similar test days) rather than the unconditional marginal. Agreement with the conditional reference confirms effective use of the FP signal.

Per-Farm Coverage. The proportion of real FE observations that fall inside the Q5–Q95 generated prediction interval, with a 90% nominal target. Shortfalls indicate excessively narrow (overconfident) intervals; exceedances indicate intervals that are too wide to be informative.

Inter-Hour Correlation Heatmap. The 24×24 Pearson correlation matrix across FE hours. Authentic wind FE exhibits a warm banded pattern — nearby hours strongly positively correlated, with correlation diminishing as separation grows. The mean absolute discrepancy $|\Delta|$ between the real and generated matrices quantifies the severity of any temporal-coupling failure.

9 Results and Discussion

9.1 Summary of Quantitative Metrics

Table 4: Summary of key evaluation metrics.

Metric	Value	Target / Reference
Best Wasserstein-1 Distance	0.0290	0.02–0.05 (good range)
Mean CRPS (all farms, all hours)	0.0554	Lower is better
CRPS Std across 20 sets	2.3×10^{-5}	Low $\Rightarrow$ training stability
Mean QQ R^2	0.379	$\rightarrow 1.0$ (ideal)
Mean 90% PI Coverage	78.2%	$\geq 90\%$ (target)
Mean Inter-Hour Corr. $ \Delta $	0.533	$\rightarrow 0$ (ideal)

9.2 Training Convergence

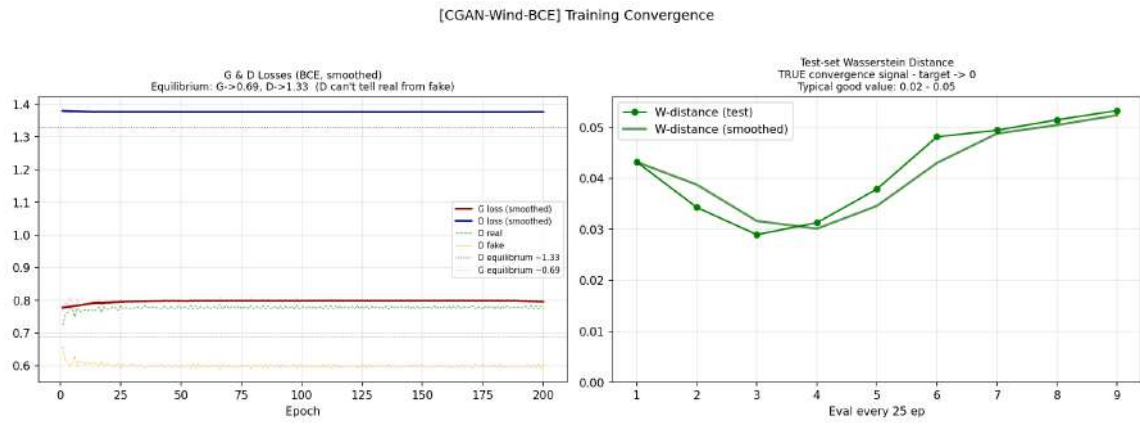


Figure 1: Training convergence curves.

9.3 Hourly CDF Comparison — Farm 0

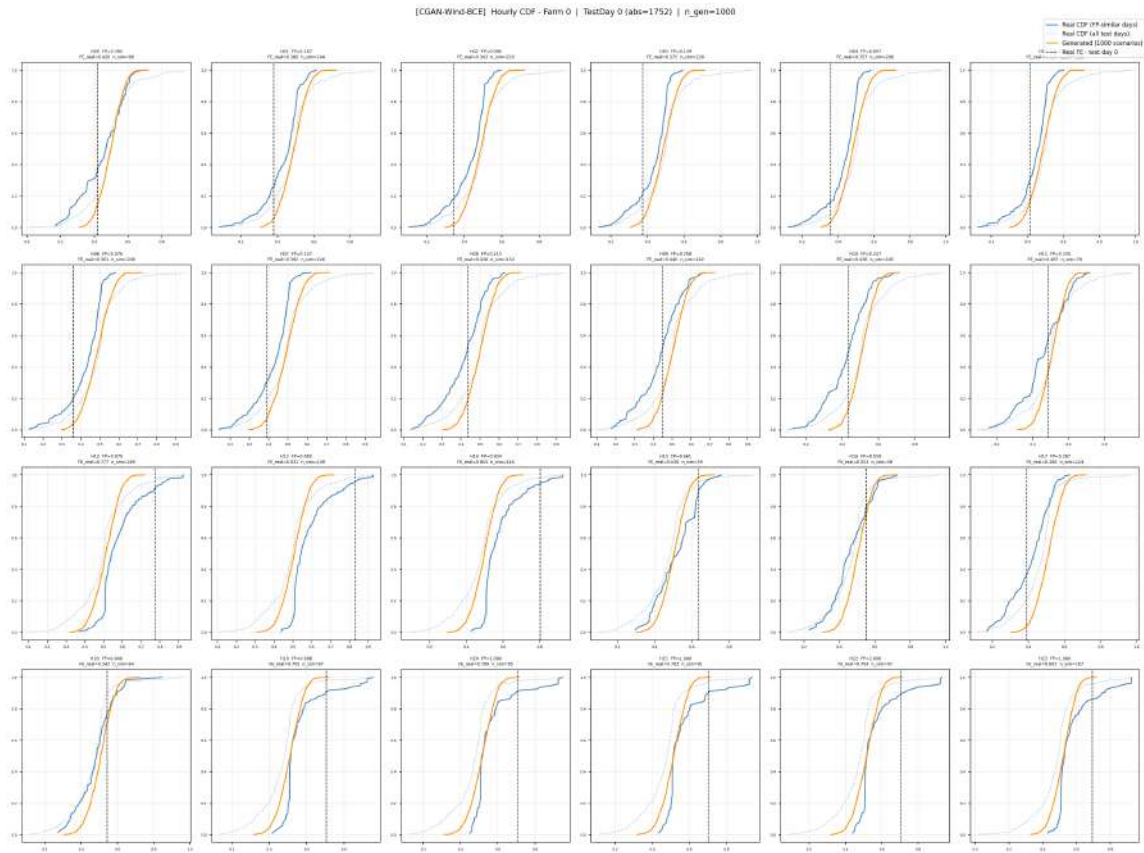


Figure 2: Per-hour CDF comparison for Farm 0 over all 24 hours.

9.4 All-Farm CDF at Hour 12

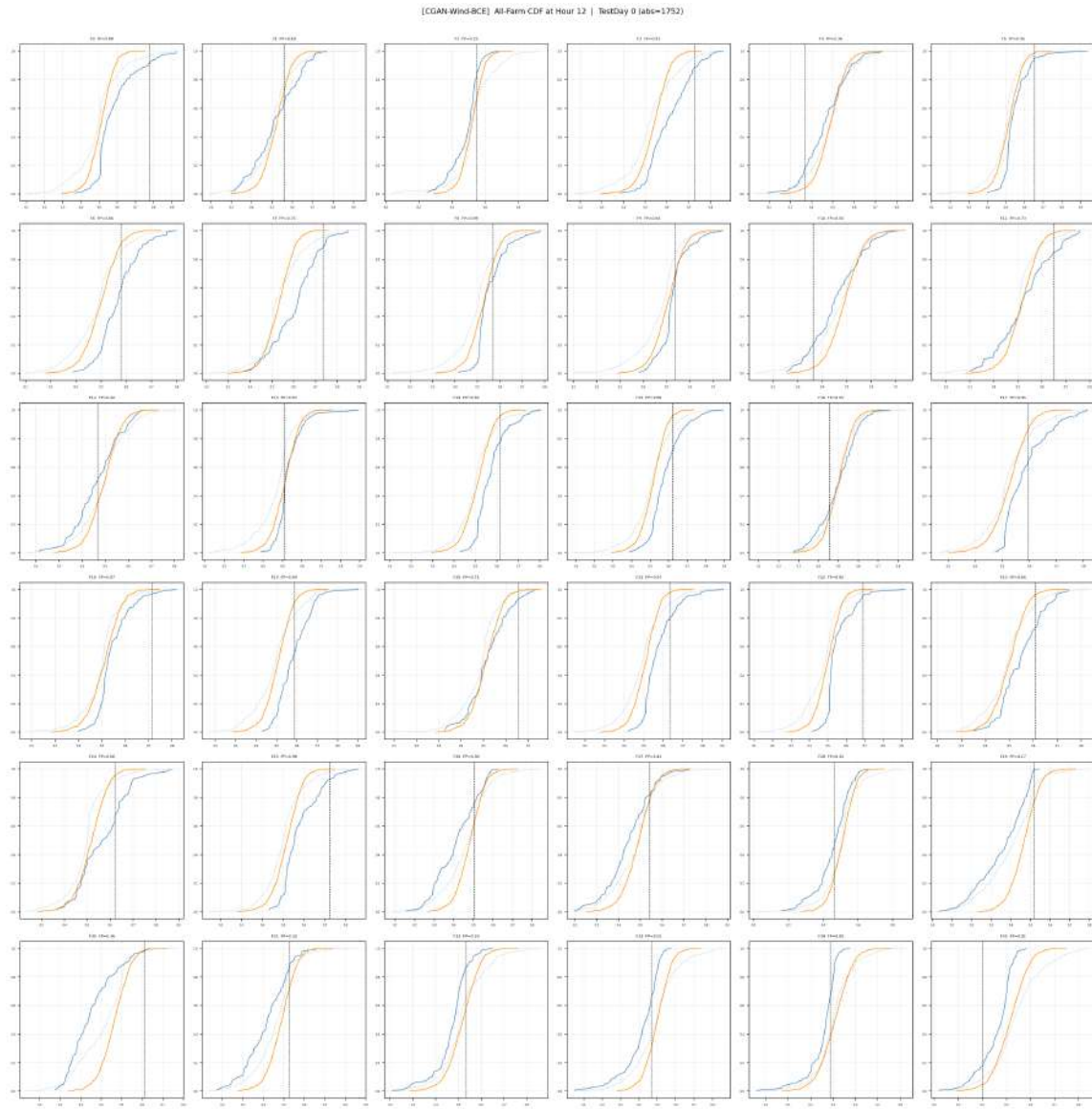


Figure 3: Marginal CDF at hour 12 across all 36 farms on the first test day.

[CGAN-Wind-BCE] hourly QQ - farm 0 | TestDay 0 (obs=1752)
X-axis: quantiles | Y-axis: quantiles (P-similar days) | Mean R2=0.378

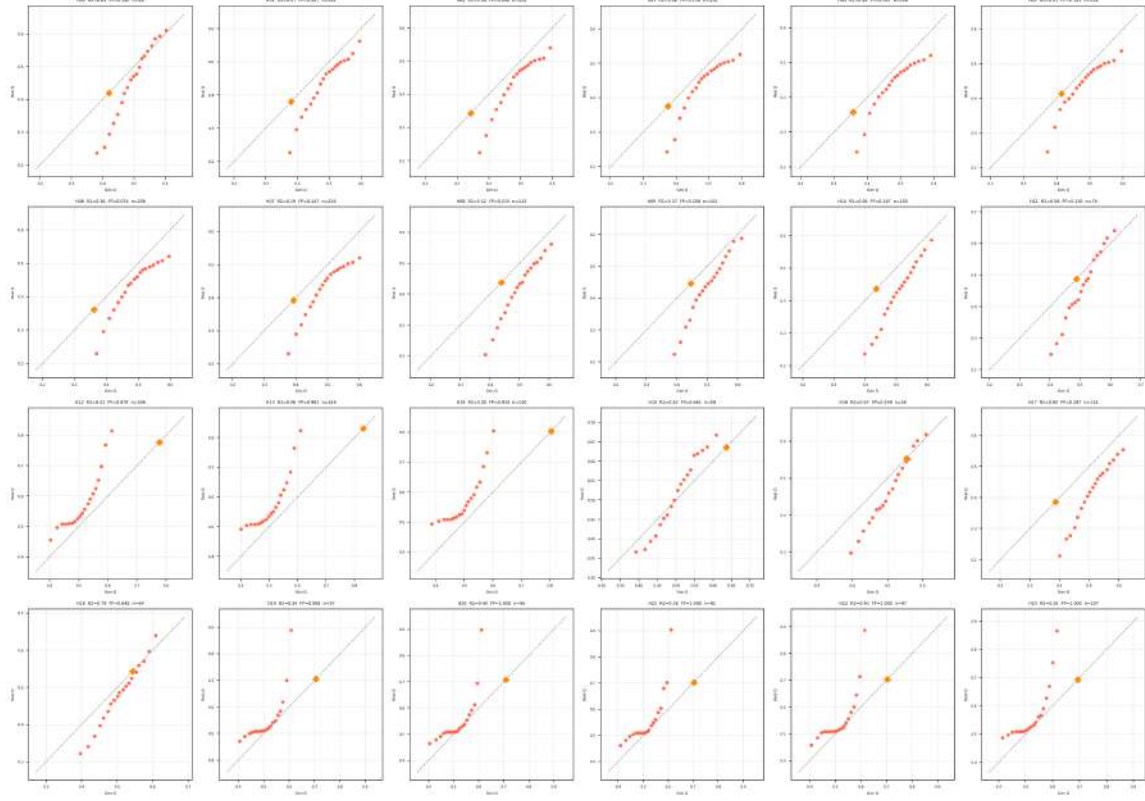


Figure 4: Quantile–quantile plots for all 24 hours of Farm 0.

9.5 CRPS per Farm per Hour

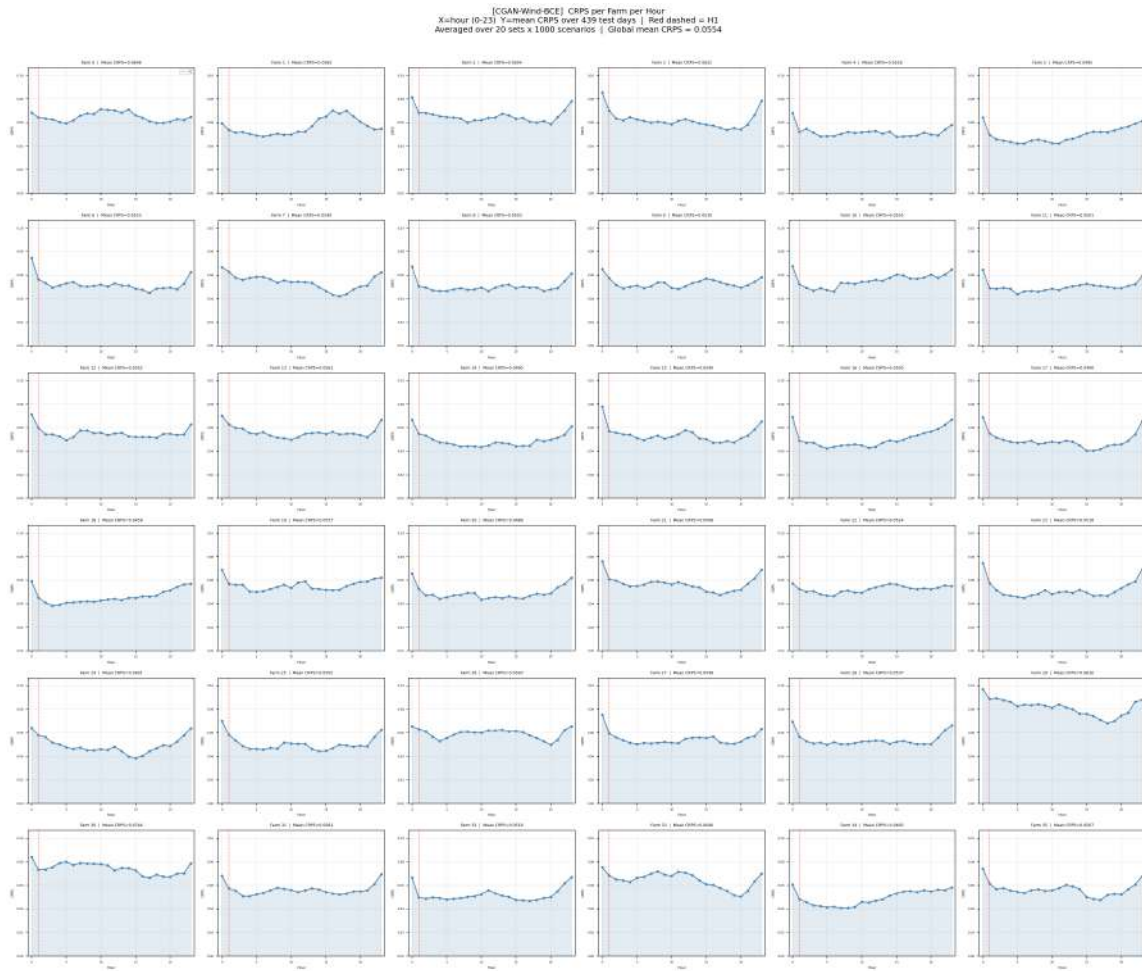


Figure 5: Hourly CRPS profiles for all 36 farms.

Figure 6: CRPS at a randomly selected hour (H1) across all 36 wind farms. Green bars indicate lower (better) CRPS; red bars indicate higher (worse) CRPS. The horizontal dashed line marks the cross-farm mean.

9.6 Fan Chart — Farm 0

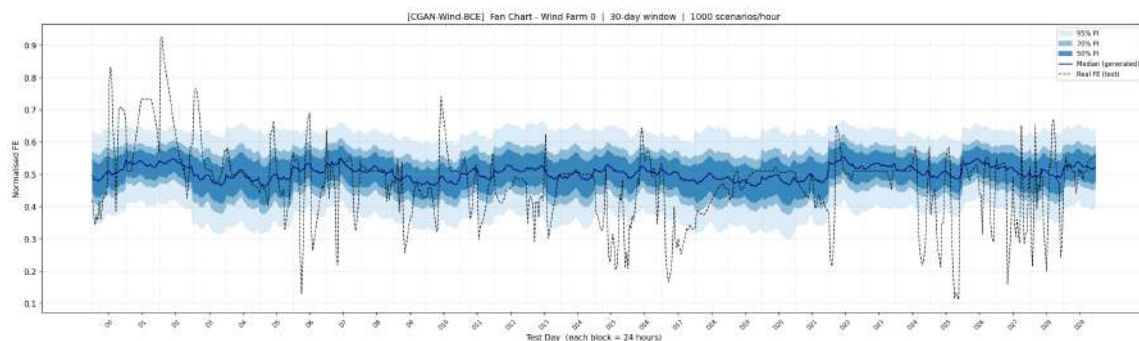


Figure 7: Fan chart for Farm 0 spanning

9.7 Inter-Hour Correlation Heatmap

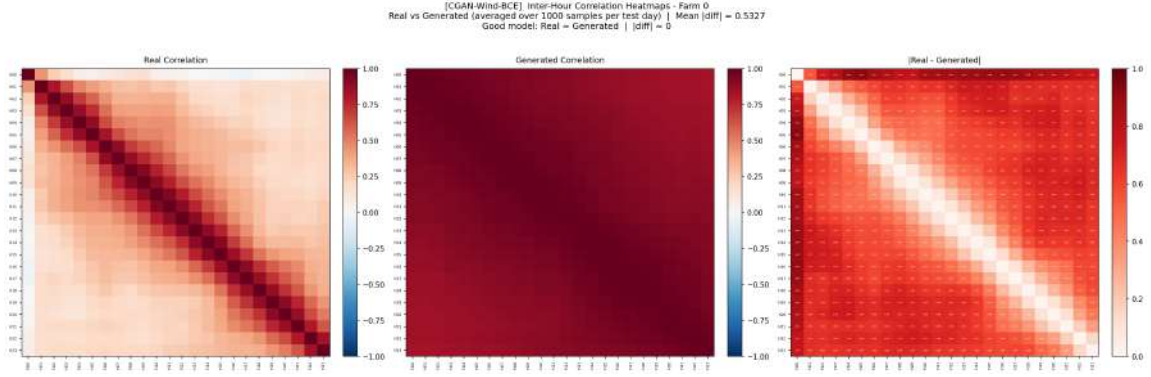


Figure 8: Inter-hour Pearson correlation matrices for Farm 0.

10 Challenges and Limitations

Variance collapse with flat conditioning. An earlier model version using a flat 864-dimensional FP condition exhibited severe variance underestimation, with generated CDFs collapsing into narrow steep curves. Diagnosing this and redesigning the conditioning into the three-tier structured form required multiple iterations. The structured pathway substantially mitigated — but did not entirely eliminate — this failure mode, as evidenced by the residual coverage shortfall.

Temporal over-coupling. Inter-hour correlation $|\Delta| = 0.533$ is the weakest reported metric. The shared per-cell MLP has no architectural mechanism for capturing intra-day persistence (no recurrent, attention, or convolutional component along the hour axis), so all 24 hours of a generated profile end up near-identically distributed.

Systematic under-coverage. A mean 90% PI coverage of 78.2% falls 11.8 percentage points short of the nominal target. The adversarial BCE loss provides no calibration-aware signal, leaving the generator without a direct incentive to broaden intervals where necessary.

Heterogeneity across farms. Farms F29 and F30 display CRPS roughly twice the global mean and coverage below 75%. A single shared generator trained simultaneously on all farms must average over heterogeneous distributional shapes; statistical outliers therefore receive miscalibrated outputs.

Limited training data. Approximately 1,752 training days are used to learn an effective 864-dimensional joint distribution, limiting the achievable fidelity of any high-capacity model.

No statistical baseline. Time constraints prevented benchmarking against classical scenario-generation methods (Gaussian Copula, ARMA + Bootstrap, Quantile Regression). The current results are therefore self-referential rather than comparative — a limitation directly addressed in future work.

11 Future Work

1. **Statistical baseline comparison.** Benchmark the proposed CGAN against established methods (Gaussian Copula, ARMA + Bootstrap, Quantile Regression) to quantify the deep-generative advantage on this dataset.
2. **Loss-function upgrade.** Replace BCE with the Wasserstein-GP critic or hinge loss, both of which have demonstrated stronger coverage and tail fidelity in published wind-scenario CGAN work.
3. **Architectural enhancements.**
 - Introduce convolutional layers (CNN-MLP / Conv2D) over the (24×36) FE map to better capture joint spatial-temporal structure.
 - Add an attention or transformer block over the 24 hours to recover the diagonal-band inter-hour correlation structure.
4. **Diversity regularisation.** Add explicit anti-collapse terms (mini-batch standard deviation, PacGAN-style packing, or feature-matching loss) to push 90% PI coverage above the nominal target.
5. **Calibration-aware penalty.** Add an interval score or pinball loss evaluated at multiple quantile levels to create a direct training incentive for correctly-sized prediction intervals.
6. **Farm-specific embeddings.** Introduce learnable farm-embedding vectors so that outlier farms (F29, F30) can develop distinct distributional behaviour within the shared generator.
7. **Time-series and modern variants.** Compare against TimeGAN, DoppelGANger, and conditional diffusion models, which have recently demonstrated stronger temporal-correlation fidelity for renewable scenario generation.
8. **End-use validation.** Evaluate generated scenarios in a downstream stochastic unit-commitment or reserve-sizing simulation to assess practical utility beyond statistical metrics.

12 Conclusion

A Conditional GAN with a novel three-tier structured conditioning pathway has been developed, trained, and rigorously evaluated for probabilistic wind-power forecast-error generation across 36 spatially-correlated wind farms using the NREL WIND Toolkit. Each daily sample is structured as 24×36 to preserve cross-farm spatial coupling within every scenario.

The model achieves stable adversarial convergence with a best Wasserstein-1 distance of 0.029 (within the typical 0.02–0.05 “good” range) and a mean CRPS of 0.055. The CRPS standard deviation across 20 independent ensembles is 2.3×10^{-5} , two orders of magnitude smaller than

the mean CRPS, confirming that the structured-conditioning design successfully eliminated the run-to-run variance-collapse failure mode observed in earlier flat-conditioning iterations.

However, the evaluation also reveals clear residual limitations. Mean 90% prediction-interval coverage reaches only 78.2% against the 90% target, indicating systematic under-dispersion. Inter-hour correlation $|\Delta| = 0.533$, with generated correlations uniformly saturated near 1.0, indicating that the model fails to reproduce the diagonal-band intra-day persistence structure of real wind FE. Mean QQ R^2 of 0.379 shows that central distributions align reasonably but tails remain misaligned.

The per-cell MLP decoder’s limited capacity for joint spatial–temporal structure, the BCE adversarial objective, and the modest $\sim 1,752$ -day training set jointly constrain the achievable performance. This work establishes a working CGAN baseline for multi-farm probabilistic scenario generation and points clearly toward convolutional or attention-based decoders, Wasserstein-GP losses, explicit diversity regularisation, and statistical-method benchmarking as the next steps. Source code is accessible at: https://drive.google.com/drive/folders/1MOT2sDQrpEcfZ50Bp3hplhxlqh4C6WKC?usp=drive_link.

References

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